

Magnetic Correlations and Superconducting Pairing in an Interaction-Driven Altermagnetic Hubbard Model

Summary. Our group has three published works on altermagnetism (AM) in which the spin splitting is **imposed** by a spin-dependent hopping. This project studies the complementary, and to our knowledge unexplored, problem: a fully **spin-independent** Hubbard model on the anisotropic checkerboard lattice, in which altermagnetism can only be **generated spontaneously by the interaction**. Two results anchor the plan. (i) A symmetry analysis shows the altermagnetic form factor of this lattice is **dxy**, orthogonal to the **dx²-y²** channel of the square-lattice model in our pairing paper [11]; this predicts a **different dominant pairing symmetry**. (ii) Preliminary L=14 CPQMC data show Néel (π, π) is the only long-range order and it survives only near half-filling, while the doped region is magnetically short-range — precisely the regime where [11]'s mechanism predicts pairing to be enhanced. The honest verdict below is “**worth continuing, but the innovation must be narrowed to the pairing / symmetry-selection question.**”

Stage 1 — Research Context and Motivation

1.1 Research background

Altermagnets are a third class of collinear magnet, distinct from ferromagnets and conventional antiferromagnets (AFM). The two spin sublattices are related not by translation or inversion but by a **rotation**, which forbids a net moment yet produces an even-parity, momentum-dependent spin splitting of the bands with a definite d-, g- or i-wave symmetry [1–5]. This unites the zero stray field and THz dynamics of an antiferromagnet with the spin-split, spin-polarized transport of a ferromagnet, without spin-orbit coupling [9,10].

Two **physically distinct routes** to altermagnetism must be separated:

- **Imposed (single-particle) AM:** a spin-dependent hopping is written into the Hamiltonian, so the spin splitting exists already at $U = 0$. This is the route of our group's three published works, including the pairing study [11], where the anisotropy t_A acts oppositely on the two spins.
- **Interaction-driven (emergent) AM:** the Hamiltonian is fully spin-symmetric; the altermagnetic splitting appears only after a magnetic moment forms spontaneously through the interaction. Proposed from itinerant magnetism [12], orbital order [13], and on modified Lieb [14] and Shastry–Sutherland [15] lattices; correlations can also generate an altermagnetic anomalous Hall response [16].

The essential gap: nearly every interaction-driven study to date is **mean-field or slave-boson**. Unbiased many-body treatments are rare, and the **superconducting pairing problem has never been addressed in a spontaneous altermagnet by an unbiased method**. This project targets exactly that intersection.

1.2 Core scientific problem (one sentence)

Does altermagnetism that is **generated by the interaction** — rather than imposed at the single-particle level — reorganize the magnetic correlations and the superconducting pairing of a single-band Hubbard model, and does the symmetry of the emergent order (here dxy) select a pairing channel different from the imposed square-lattice case (dx²-y²)?

1.3 Why it is worth studying — concretely

- **Basic mechanism:** tests whether the RVB-like “pairing where AFM is suppressed” mechanism of [11] is a property of altermagnetism itself, or an artifact of imposing the spin splitting. These give opposite conceptual pictures.
- **Theory link:** supplies the first numerically-unbiased benchmark for the many mean-field predictions of interaction-driven AM [12–15], where correlations are exactly what mean field cannot capture.
- **Materials / experiment link:** checkerboard / planar-pyrochlore motifs occur in real compounds; a symmetry-selected pairing channel is experimentally falsifiable (spin-split ARPES, phase-sensitive Josephson probes).
- **New knowledge if solved:** a design principle — lattice geometry fixes the altermagnetic symmetry channel, and the pairing follows it — rather than a single-model result.
- **Missing link if not studied:** we would still not know whether “altermagnetic superconductivity” requires the splitting to be imposed, or emerges from pure repulsion plus geometry.

1.4 Known vs. unknown

Already resolved	Still open (long-standing)
Symmetry classification of AM and its d/g/i-wave order [1,4,5].	Whether emergent AM survives strong correlations in an unbiased ground-state calculation.
Experimental existence of AM (MnTe, RuO ₂) via Kramers-lifting and imaging [6–8].	Whether emergent AM enhances, suppresses, or reshapes superconducting pairing.
Imposed-AM square lattice: anisotropy suppresses AFM and enhances dx ² -y ² pairing [11].	Whether the pairing symmetry is selected by the lattice AM form factor (dxy vs dx ² -y ²).
Doping evolution of magnetic order in imposed-AM Hubbard (Papers 1, 2).	Whether that doping evolution changes qualitatively when the anisotropy is geometric, not spin-dependent.

1.5 Where this project plugs in

The project sits at the **junction of two established group lines**: (a) the magnetic-correlation program (Papers 1, 2) and (b) the AM-pairing program [11]. It replaces the **imposed** spin-dependent hopping with a **geometric** anisotropy δ on the checkerboard lattice, making the AM emergent, and asks the magnetic and pairing questions together. It is the natural ‘third leg’ closing the imposed-vs-emergent gap.

Stage 2 — Development Timeline of the Field

Key conceptual nodes rather than a flat citation list. ‘Limit left’ is what each node did NOT settle, which motivates the next.

Period	Node / Ref.	System & method	Core finding	Limit left
2020–22	Symmetry birth of AM [1,3–5]	Spin-group / symmetry theory	AM defined as a distinct phase; d/g/i-wave spin splitting classified	Pure symmetry; no correlation dynamics
2022–24	Experimental confirmation [6–8]	MnTe, RuO ₂ ; ARPES, imaging	Kramers-degeneracy lifting and real-space AM order observed	Weakly correlated materials, not the Hubbard regime
2024–25	Interaction-driven AM proposals [12–16]	Itinerant / orbital / Lieb / Shastry–Sutherland; mostly mean-field	AM can emerge spontaneously from interactions or geometry	Mean-field / slave-boson — correlations not captured
2024	Exact AM benchmark [16]	Kane–Mele–Hubbard; AFQMC	First unbiased AM treatment; correlation-driven Hall response	Different lattice; pairing not addressed
Group	Magnetic correlations, Papers 1–2	Square Hubbard, spin-dependent hopping; CPQMC	Competing Néel / stripe / diagonal / spiral; doping evolution	AM imposed; pairing not the focus
2026	AM pairing [11]	Square Hubbard, imposed	Anisotropy suppresses AFM and	Leaves open

		AM; CPQMC 16×16	enhances dx ₂ -y ₂ pairing (doping-free RVB analogue)	whether the splitting must be imposed
Now	This project	Checkerboard Hubbard, emergent AM; 4-method QMC	(to be determined)	(the gap we address)

2.1 Literature, classified

- **Classical / foundational:** Šmejkal–Sinova–Jungwirth PRX 12, 040501 (2022) [1]; Mazin (ed.) PRX 12, 040002 (2022) [2]; symmetry [4,5].
- **Methodological:** Scalapino–White–Zhang pair susceptibility PRB 47, 7995 (1993) [19]; determinant / constrained-path QMC PRB 40, 506 (1989) [20]; the vertex-pairing construction of [11].
- **Closest to this project:** our AM-pairing paper [11]; group Papers 1 & 2 [17,18].
- **Recent frontier:** interaction-driven AM [12–15]; exact AM benchmark [16]; experiments [6–8].
- **Contrary / cautionary:** [11] itself notes the enhancement is strongest where the AFM structure factor is diminished — a warning that a single-band emergent model, which suppresses AFM geometrically rather than by orbital exchange, may behave differently and must be checked, not assumed.

Stage 3 — Positioning of the Current Project

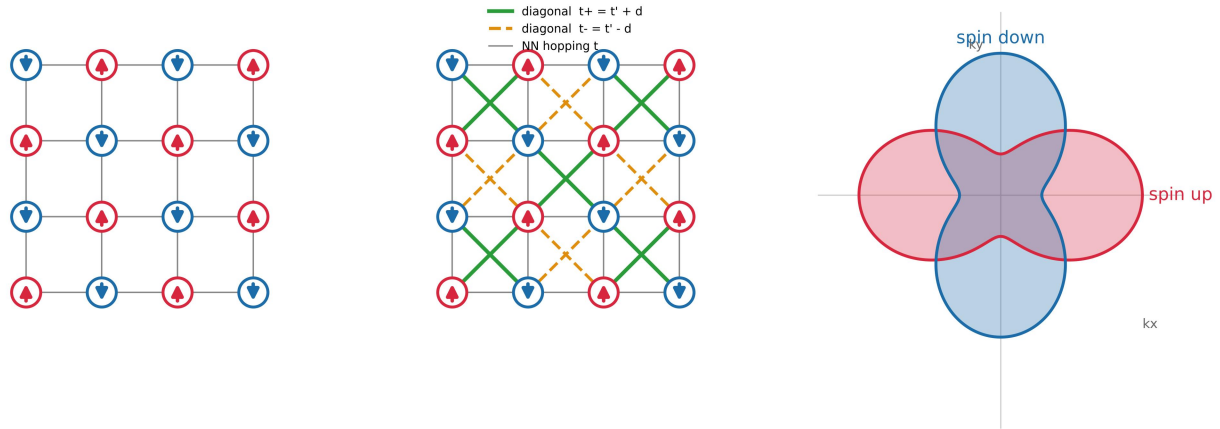


Fig. 1. From imposed to emergent AM. (a) Plain square + U : sublattices related by translation $\rightarrow$ spin-degenerate Néel (not AM). (b) Checkerboard: every plaquette crossed, half by $t_+ = t' + d$, half by $t_- = t' - d$; the two sublattices are related by a rotation $\rightarrow$ the AM condition. (c) Resulting d-wave spin splitting with zero net moment.

Dimension	Existing work [11,17,18]	This project	Real difference	Gap?
Lattice	Square	Checkerboard (planar-pyrochlore)	New geometry, $C4v \rightarrow C2v$	Partly
Hamiltonian	Spin-dependent hopping	Spin-independent; $t_{\pm} = t' \pm \delta$	AM imposed $\rightarrow$ emergent	Yes — core
AM origin	$U=0$ splitting present	$\propto M \cdot \delta$, needs a spontaneous moment	Conceptually different	Yes — core
AM symmetry	dx ₂ -y ₂	dxy	Orthogonal irreps	Yes — core
Parameter axes	(t_A , t' , n)	(δ , n) at $U=4$; $U=0-4$	Geometric knob δ	Partly
Method	CPQMC	CPQMC + DQMC + CP-DQMC + ED	Adds susceptibilities + exact check	Incremental
Observables	$S(q)$, $\Delta n(k)$, vertex pairing	+ symmetry-resolved AM weight, χ_s , pair susceptibility	New AM-symmetry probe	Yes
Objective	Show imposed AM enhances d-wave	Test if emergent AM does, and which channel	Mechanism, not phenomenon	Yes
Focus	Phenomenon + mechanism	Mechanism + design principle	Symmetry selection	Yes

Done before?	No unbiased emergent-AM pairing study exists	—	—	Open
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The AM-symmetry row is the single most consequential difference: dx_2-y_2 versus dx_y are orthogonal irreducible representations, and the spin-fluctuation glue that mediates pairing inherits this symmetry.

Positioning answers:

- **Where in the arc?** The ‘third leg’ unifying the group's magnetic-correlation and AM-pairing programs under emergent AM.
- **Extension, cross-over, or new?** A cross-cutting direction: it crosses the imposed-AM pairing method [11] with a geometric emergent-AM model.
- **Closest papers:** [11] (method & question), [17,18] (magnetic toolkit), [12,15] (emergent-AM concept).
- **What we really add:** (i) emergent vs imposed AM under unbiased QMC; (ii) symmetry-selection of the pairing channel; (iii) a reusable symmetry-resolved AM order parameter.
- **Already done by others:** imposed-AM pairing [11]; mean-field emergent AM [12–15]. NOT done: emergent-AM pairing, unbiased.
- **Nature of the gap:** primarily a mechanism / method gap (no unbiased treatment), secondarily an observable gap (symmetry-resolved AM weight). Not merely ‘one more parameter.’

Stage 4 — Value and Feasibility Judgment

Two independent voices, then a verdict. This section is deliberately critical: the preliminary $L=14$ data have already falsified one of our earlier claims (below), and the plan is stronger for admitting it.

Checkerboard Hubbard, $U=4$, $L=14$ (40 completed points) — Néel long-range only near half-filling; doped region short-range

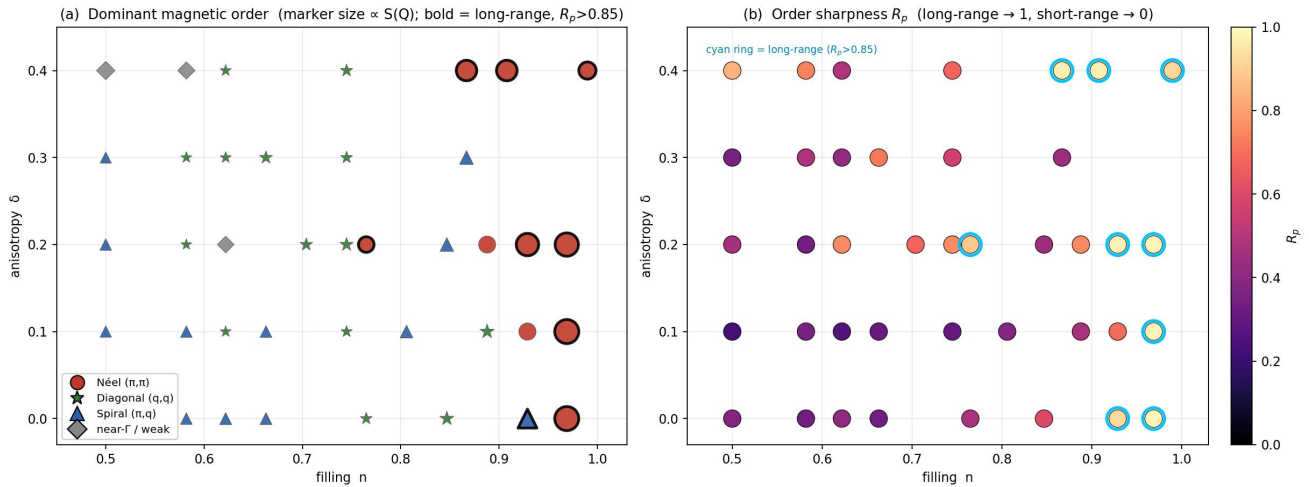


Fig. 2. Honest preliminary $L=14$ result ($U=4$). (a) Dominant order (marker size $\propto S(Q)$; bold = long-range). (b) Sharpness R_p . Only 9/40 points reach $R_p > 0.85$ and 8 are Néel near half-filling; the doped interior is short-range. The earlier $L=10$ claim of ‘ δ drives sharp diagonal order (R_p up to 0.94)’ does NOT survive at $L=14$ — the tidy $L=10$ regions were an artifact of coarse k -resolution plus forced classification.

4.1 Judgment

- **The symmetry argument (dx_y vs dx_2-y_2) is the crown jewel:** model-independent, follows from the Hamiltonian, and yields a concrete falsifiable prediction. It elevates the project from ‘same study, new lattice’ to a design principle.
- **The magnetic side is solid but modest:** Néel-only-near-half-filling with a short-range doped sea is a clean, defensible result, and it is exactly the substrate [11]’s pairing mechanism needs.

- **Four-method control (CPQMC / DQMC / CP-DQMC / ED) is a genuine asset:** ED already showed CPQMC over-orders near half-filling, so cross-checks are the credibility of the whole story.

4.2 Strict reviewer judgment

- **Magnetism alone is not novel enough:** ‘doped Hubbard is short-range, Néel near half-filling’ is expected; published on its own it reads as incremental.
- **‘Altermagnet’ must be earned, not asserted:** away from special points the emergent order may be an ordinary Néel AFM. The referee will demand $\Delta_{\text{tot}}(\delta)$ and the symmetry-resolved weight as proof, not the band cartoon.
- **Pairing-signal risk:** vertex pairing in a single-band repulsive model at $U=4$ is small; the constrained-path bias that inflates $S(\pi, \pi)$ can also bias the pairing vertex. Without ED / CP-DQMC corroboration the pairing claim is fragile.
- **Doped $S(Q)$ is small (0.2–0.9):** error bars matter; no doped-region claim without statistics.

4.3 Verdict

► **[Worth continuing, but the innovation must be narrowed].** The magnetic phase diagram is supporting material, not the headline. The project's real value is the **emergent-AM + symmetry-selected pairing** question. Re-centre the paper there.

Question	Answer
Strongest research thread	The dxy symmetry prediction — model-independent and falsifiable — that emergent AM selects a pairing channel different from the imposed square-lattice case.
Biggest novelty risk	Emergent order is just ordinary Néel AFM away from special δ , so the ‘altermagnet’ label is unearned.
Biggest technical risk	CP bias inflates $S(\pi, \pi)$ and possibly the pairing vertex near half-filling; small doped signals.
Cheapest, most informative test	Measure $\Delta_{\text{tot}}(\delta)$ and the symmetry-resolved dxy/dx ² -y ² weight at half-filling across δ — CPQMC already in hand; directly proves or kills emergent AM and its symmetry.
Continue if...	$\Delta_{\text{tot}} = 0$ at $\delta=0$ and grows $\propto M \cdot \delta$ with dxy dominant; and a pairing-vertex dome appears where AFM is suppressed.
Adjust / stop if...	dxy and dx ² -y ² weights are comparable (no clean selection), or the pairing vertex shows no δ -trend beyond error bars.

Stage 5 — Key Observable Design

Observables chosen by the questions and graded by role: core (answer the main question), diagnostic (separate rival explanations), auxiliary (aid understanding), control (exclude finite-size / boundary / numerical artifacts).

5.1 Core observables

- **Spin structure factor $S_z(\mathbf{q})$** = $(1/N) \sum_{ij} e^{i\mathbf{q} \cdot \mathbf{r}} \langle S_z(i) S_z(j) \rangle$; dominant order from $Q = \text{argmax}$. Answers ‘which order.’ Confound: CP over-ordering inflates the amplitude near half-filling $\rightarrow$ trust Q , not $S(Q)$. From CPQMC (dir-kVals/sdwz.dat).
- **Sharpness R_p** = $[S(Q) - \langle S \rangle_{\text{shell}}] / S(Q)$; an amplitude-insensitive peak-shape measure $\rightarrow$ robust where CP fails. $R_p \rightarrow 1$ long-range, $\rightarrow 0$ short-range. This is the observable that exposed the $L=10$ artifact.
- **Symmetry-resolved AM weight Δ_Γ** = $\sum_k \Delta n(k) f_\Gamma(k)$, with $f_{\text{dxy}} = \sin k_x \sin k_y$, $f_{\text{dx}^2\text{-y}^2} = \cos k_x - \cos k_y$, $\Delta n = n \uparrow(k) - n \downarrow(k)$. The decisive test of the symmetry claim; expected dxy-dominant here, dx²-y² in [11]. From CPQMC momentum occupations.

- **Vertex (connected) pairing function** $N_Vertex, \zeta(k)$ [connected part of $\langle \Delta_{\zeta} \Delta_{\zeta}^{\dagger} \rangle$, Eq. 3 of [11]], $\zeta = dx, dy, dx^2 - y^2, \text{ext-s}, p \uparrow \downarrow$, with checkerboard bond form factors. Positive vertex = genuine interaction-driven pairing.

5.2 Diagnostic observables

- **AM order parameter** $\Delta_{\text{tot}} = \sum_k |n_{\uparrow}(k) - n_{\downarrow}(k)|$ with $\sum_k \Delta n = 0$: proves compensated AM; must vanish at $\delta=0$.
- **Spin susceptibility** $\chi_s(q) = \int_0^{\beta} d\tau \langle S_z(q, \tau) S_z(-q, 0) \rangle$: finite-T instability; separates true ordering from a broad equal-time peak. From DQMC / CP-DQMC.
- **Pair-field susceptibility** P_{ζ} vs its bare bubble P_{ζ}^0 : $P_{\zeta} > P_{\zeta}^0$ is the standard attractive-channel criterion; the τ -integration weights low-energy pairs the equal-time vertex misses.

5.3 Auxiliary and control

- **Auxiliary**: real-space $C_SDW(r)$ (decay = long/short range); non-interacting Fermi surface / DOS / van Hove vs δ (single-particle ‘why’); double occupancy and energy (thermodynamic consistency).
- **Control**: finite-size scaling $L = 10, 14, 16$; closed-shell fillings only (unique trial WF); ED cross-check on small clusters; CPQMC vs CP-DQMC to bound the constrained-path bias.

5.4 Scientific-question ↔ observable map

Scientific question	Core observable	Physical meaning	Expected signal	Confounds
Is the AM emergent?	$\Delta_{\text{tot}}(\delta), \chi_s$	compensated spin split from U	= 0 at $\delta=0$, grows $\propto M\delta$	ordinary Néel mimics a finite moment
Which symmetry channel?	$\Delta_{dx}, \Delta_{dx^2-y^2}$	irrep of the spin split	$dx \gg dx^2 - y^2$	mixed if C4 weakly broken
Where is order long-range?	$Q, R_p, C_SDW(r)$	order vs fluctuation	Néel $R_p \rightarrow 1$ near $n=1$; doped $\rightarrow 0$	CP inflates $S(Q)$ near $n=1$
Is pairing enhanced?	$N_Vertex, \zeta, P_{\zeta}$	connected pair correlation	dome vs δ in the dx channel	ext-s background; CP bias
Which pairing symmetry?	ratio N_d / N_s	competition of channels	dx leads, rising with δ and U	ext-s background; finite size

Stage 6 — Why the Observables Change (causal chains)

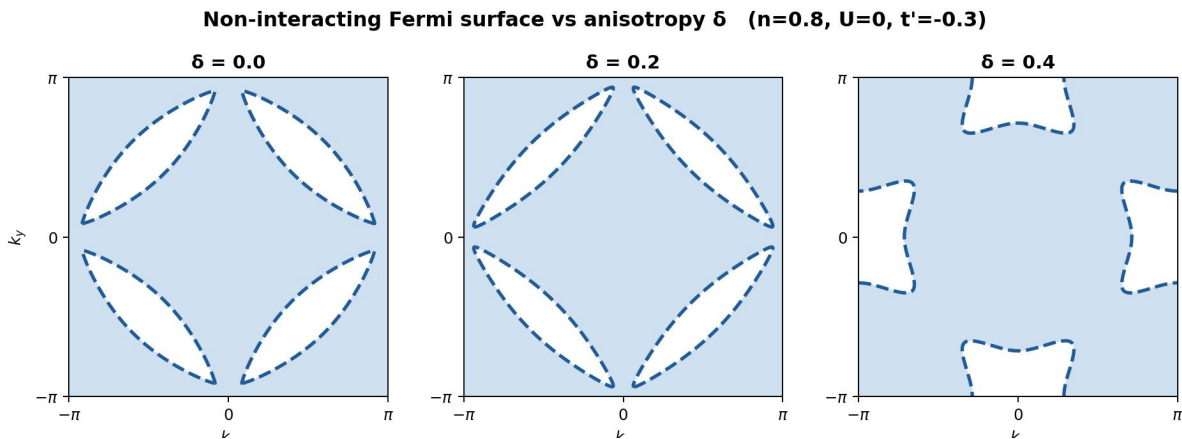


Fig. 3. Non-interacting Fermi surface vs δ . Increasing δ drives a Lifshitz reconstruction that destroys (π, π) nesting.

Master causal chain:

$\delta \uparrow \rightarrow$ the diagonal term $m(k) = 4\delta \sin k_x \sin k_y$ grows in $H \rightarrow C4v \rightarrow C2v$, and the $U=0$ Fermi surface undergoes a Lifshitz reconstruction (Fig. 3), destroying (π, π) nesting $\rightarrow$ the AFM instability weakens while a dxy-symmetric particle-hole channel opens $\rightarrow$ the spin background reorganizes from Néel toward short-range / diagonal fluctuations $\rightarrow S_z(q), R_p, \Delta n(k)$ and the pairing vertex all change.

Change-by-change reading:

- **Weaken (AFM):** $N_{S_z}(\pi, \pi)$ and its R_p fall as δ breaks nesting — symmetry-allowed, robust.
- **Peak move (incommensuration):** Q drifts off (π, π) on doping; filling-dependent — model/parameter-dependent, not universal.
- **Non-monotonic (pairing):** the vertex may rise then fall with δ (a dome): too little δ keeps AFM dominant, too much destroys coherence — direction set by competing energies, so we do NOT pre-commit to a sign.
- **Symmetry change (AM onset):** Δ_{tot} turns on from 0 as $M \cdot \delta$ — fixed by symmetry (must vanish at $\delta=0$).

For each: the direct explanation (nesting loss $\rightarrow$ AFM suppression) is symmetry-backed; alternatives (e.g. finite-size peak wandering) are separated by R_p , by $\chi_s(q)$, and by $L = 10/14/16$ scaling.

Stage 7 — The New Physical Condition (δ) and Its Effects

- **Hamiltonian term:** δ enters only through the diagonal hopping $t_{\pm} = t' \pm \delta$ on the crossed plaquettes; in the 2-sublattice Bloch form it is $m(k) = 4\delta \sin k_x \sin k_y$.
- **Directly coupled to:** the inter-sublattice (A/B) kinetic channel — spin-independently.
- **Symmetry / selection rules:** lowers $C4v \rightarrow C2v$; makes Néel order altermagnetic with a dxy form factor; the spin split is forbidden at first order in δ alone (needs M): $\Delta E \propto M \cdot \delta$.
- **Energy scales / competition:** δ tunes diagonal kinetic energy against U -driven superexchange and against NN kinetic energy — the knob balancing AFM nesting vs diagonal correlation vs pairing.
- **Preferential effect:** first on Fermi-surface topology (Lifshitz, Fig. 3) and $\Delta n(k)$; then on $S_z(q)$ and the pairing vertex.
- **Order in perturbation:** the AM splitting is second order ($\propto M \cdot \delta$, needs a moment); the FS distortion is first order in δ ; a net moment is symmetry-forbidden.
- **Strong- δ regime:** crossings likely — $\delta = |t'| = 0.3$ switches t_+ off (connectivity change); $\delta > |t'|$ flips the sign of t_+ (maximal AM contrast). State reconstruction expected across this window.
- **Artifacts to watch:** open-shell fillings (use closed shells); CP over-ordering near $n=1$; k -resolution at fixed L (the $L=10 \rightarrow L=14$ lesson); boundary effects $\rightarrow$ finite-size scaling.

7.1 Predicted effect on each key observable

Observable	Possible change	Direct mechanism	Alternative	Supporting / excluding signal
Δ_{tot}	$0 \rightarrow$ finite as $\delta \uparrow$	emergent AM, $\propto M \cdot \delta$	ordinary Néel (also finite moment)	symmetry-resolved dxy weight distinguishes
Δ_{dxy}	dominant, grows with δ	dxy form factor of lattice	dx^2-y^2 leakage if $C4$ weakly broken	compare $\Delta_{dx^2-y^2}$; must be much smaller
$N_{S_z}(\pi, \pi)$	suppressed with δ	Lifshitz loss of nesting	CP bias masking	cross-check χ_s , CP-DQMC, ED
Q (doped)	drifts off (π, π)	incommensurate response	finite-size peak wander	R_p , L -scaling
$N_{Vertex}(d)$	dome vs δ (non-monotonic)	AFM suppression $\rightarrow$ RVB pairing	ext-s background; CP bias	$P_{\zeta} > P_{\bar{\zeta}}$; channel ratio; ED

Direction-of-change honesty:

- **Model-independent (symmetry):** $\Delta_{\text{tot}} = 0$ at $\delta=0$; the AM form factor is d_{xy} ; a net moment is forbidden.
- **Parameter-dependent:** the doped ordering wavevector Q ; the exact δ of any crossover; absolute pairing magnitudes.
- **Likely non-monotonic:** the pairing vertex vs δ (dome). We do not force a sign.
- **Key competing energy:** diagonal kinetic energy ($\propto \delta$) vs superexchange ($\propto t^2/U$) vs NN kinetic energy — their balance sets every direction of change.

Stage 8 — Synthesis and Priorities

1. **Scientific background.** Altermagnetism as a third collinear-magnet class; the imposed-vs-emergent divide; pairing in a spontaneous AM is untouched by unbiased methods.
 2. **Why worth studying.** Tests whether [11]'s AFM-suppression $\rightarrow$ d-wave mechanism is intrinsic to AM or an artifact of imposing the splitting; first unbiased benchmark for emergent-AM theory.
 3. **Timeline.** Symmetry birth (2020–22) $\rightarrow$ experiments (2022–24) $\rightarrow$ emergent-AM proposals, mostly mean-field (2024–25) $\rightarrow$ exact AM benchmark [16] $\rightarrow$ group imposed-AM pairing [11] $\rightarrow$ this emergent-AM project.
 4. **Closest work.** [11] (method + question), Papers 1–2 (magnetic toolkit), [12,15] (emergent-AM concept).
 5. **Positioning.** Cross-cutting: imposed-AM pairing method $\times$ geometric emergent-AM model; the ‘third leg’ closing the imposed-vs-emergent gap.
 6. **Real innovation.** (i) emergent vs imposed AM under unbiased QMC; (ii) symmetry-selected pairing channel (d_{xy} vs $d_{x^2-y^2}$); (iii) a reusable symmetry-resolved AM order parameter.
 7. **Main novelty risk.** Emergent order may be ordinary Néel away from special δ — ‘altermagnet’ must be proven by $\Delta_{\text{tot}}(\delta)$ plus the symmetry weight.
 8. **Is it really useful?** Yes, if narrowed to the pairing / symmetry question. A negative result (emergent AM does NOT enhance pairing) is itself publishable: it would show [11]'s effect needs the imposed splitting.
 9. **Core observables.** $\Delta_{\text{tot}}(\delta)$, $\Delta_{d_{xy}}/\Delta_{d_{x^2-y^2}}$, Q and R_p , vertex pairing N_ζ and pair susceptibility P_ζ , $\chi_s(q)$; with ED / CP-DQMC controls.
 10. **Observable $\rightarrow$ physics.** $\delta \rightarrow$ Lifshitz loss of nesting $\rightarrow$ AFM suppression + d_{xy} channel opening $\rightarrow$ changes in $S(q)$, R_p , $\Delta n(k)$, pairing vertex.
 11. **Effect of δ .** First-order FS distortion; second-order ($\propto M \cdot \delta$) AM splitting; crossover / state reconstruction expected near $\delta = |t'| = 0.3$ and $\delta > |t'|$.
 12. **Most likely mechanism.** Geometric anisotropy suppresses (π, π) nesting; where Néel weakens, a d_{xy} pairing channel is favoured — an emergent-AM analogue of the doping-free RVB route of [11].
 13. **Undetermined now.** Whether the pairing vertex forms a δ -dome; the true doped-region order beyond error bars; the magnitude of CP bias on the pairing vertex.
 14. **Highest-priority next step.** At half-filling (the 5 δ points now archived), sweep δ and measure Δ_{tot} and $\Delta_{d_{xy}}/\Delta_{d_{x^2-y^2}}$ — the single cheapest test that proves emergent AM and its symmetry; then the vertex-pairing dome vs δ , cross-checked by CP-DQMC / ED.
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Bottom line. Re-centre the project on emergent altermagnetism and symmetry-selected pairing. The magnetic phase diagram (Néel near half-filling, short-range when doped) is honest supporting material, not the headline. The decisive, cheap first experiment is $\Delta_{\text{tot}}(\delta)$ with the dx_y/dx^2-y^2 decomposition at half-filling; the headline experiment is the vertex-pairing dome vs δ in the dx_y channel, controlled against ED and CP-DQMC.

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